

HealthConnect Clinic

Week 7 Project Summary — Data Analytics Track

1. What I planned to test

Whether the Week 6 statistical outputs (driver validation, the compound risk score, and the reminder-by-risk-tier finding) were correct and reproducible, whether the risk-score gradient held across patient segments, and whether a more powerful statistical test would change the Week 6 "suggestive" reminder-by-risk conclusion.

2. What I actually tested

Reproduced every Week 6 KPI and statistic from the raw dataset; audited the dataset for data-quality issues; tested the risk-score gradient across gender, age group and appointment type; replaced Week 6's two split-sample reminder tests with a single, more powerful pooled interaction model; and validated the risk score against an independent model-based benchmark.

3. Most important testing results

All Week 6 statistics reproduced exactly except one: the reported "risk score ≥ 2 " no-show rate (59.0%) was an unweighted average rather than the correct n-weighted pooled rate (56.0%). The risk-score gradient held in 12 of 13 tested segments. The reminder-by-risk finding, retested with a properly powered interaction model on the full sample ($p = 0.732$), is now confirmed rather than merely suggestive.

4. Issues or weaknesses identified

An unweighted-average error in one headline KPI; an under-powered split-sample test used in Week 6 for the reminder-by-risk question.

5. Improvements / refinements made

Corrected the headline no-show rate to 56.0%; replaced the split-sample reminder test with a single pooled interaction model; rebuilt the dashboard to show the correction and the new tests; closed the cross-track dependency by incorporating Data Science's actual Week 7 refined model and held-out benchmark once received.

6. Retesting results

The corrected 56.0% figure, the segment-stability check, and the pooled interaction test all reproduce consistently on repeated runs of the notebook.

7-9. Cross-track collaboration

Track collaborated with: Data Science. **What was tested collaboratively:** the compound risk score was compared against Data Science's actual Week 7 refined model (a regularized Logistic Regression, 5 features), evaluated on their held-out test set. **What changed as a result:** the original in-sample benchmark in this report ($r=0.658$, AUC 0.609/0.680) was superseded by the more reliable held-out figures Data Science provided ($r=0.642$, AUC 0.596/0.662); this cross-track dependency is now closed.

10. Key findings / validation outcomes

The Week 6 risk score and driver findings are substantively correct and now more rigorously validated: reproducible, segment-stable, and directionally consistent with an independent model. One headline figure required correction. The reminder effect not varying by risk tier is now a confirmed, adequately powered result.

11. Important decisions made

Decided to correct and use the pooled (56.0%) rate going forward rather than the originally reported figure; used an interim in-house model for cross-track validation while waiting on Data Science, then replaced it with their actual model once received, and adopted their held-out AUC figures as the more reliable reference over this report's original in-sample figures.

12. Remaining limitations

Dataset remains fictional/synthetic; findings are observational (association, not causation); this report's original in-sample benchmark is retained for transparency but superseded by Data Science's held-out figures; risk-score thresholds remain heuristic; the reminder-timing/channel A/B test is designed but not yet run.

13. Remaining issues or dependencies

The minor non-monotonic dip in the 25-34 age group should be monitored on updated data; the stratified reminder-timing test needs to be executed in Week 8; the held-out AUC benchmark should be propagated to any other project document that still cites the original in-sample figures.

14. Contribution to the overall HealthConnect project

This testing cycle gives the HC-POD a statistically validated, segment-stable, and now correctly reported risk-triage tool, plus adequately powered evidence that reminders should stay universal while additional support is targeted by risk tier — and, with Data Science's results incorporated, a closed-loop cross-track validation ready for Week 8 final integration.

15. What must be completed before Week 8

Design and, where feasible, execute the stratified reminder-timing/channel test; propagate the corrected 56.0% figure and the held-out AUC benchmark (0.596/0.662) to any other project document or dashboard; coordinate with Data Science and ML Engineering on how the rule-based score and the refined model will co-exist operationally.